

ANIRUDH DIWAKAR

+1 (602) 849-3715 | diwakar.anirudh1@gmail.com | Tempe, AZ, USA | linkedin.com/in/anidif22

SKILLS & TOOLS

CAD: SolidWorks, Fusion 360, CURA, Solid & Surface modelling, 3DEXperience CATIA

CAE: Ansys Fluent, Ansys mechanical, OpenFOAM, SolidWorks CFD, Salome, ABAQUS

Fundamentals: Fluid Mechanics, Stress Analysis, FEM/FEA, CFD

Supplementary: MATLAB, Python, Linux/Ubuntu, Windows

CERTIFICATIONS

ENGR2000X: A Hands-on Introduction to Engineering Simulations | <https://courses.edx.org/certificates/c40f5e0f1e29414ab99df002c4834adb>
ANSYS Fluent and Mechanical Hands-on Training| Rocket Nozzle Flange Analysis| Laminar and Turbulent Airfoil| Fluid-Structure interaction

PROFESSIONAL EXPERIENCE

PurcellAI

Tempe, AZ, USA

Mechanical Engineer- CFD

November 2024 - Present

- Co-developed smart inhaler device and optimized particle dispersion model for improved medication delivery by 20%
- Redesigned and modified InhalerPro device for easy hardware assembly, reducing 3D printing time by 10%
- Streamlined CFD-DEM approach and model acquisition to maintain project focus and enhance efficiency
- Optimized OpenFOAM simulations through Boundary condition control, ensuring solver compatibility of IcoFOAM files
- Implemented robust version control for models using SolidWorks PDM and project management boards (Trello)

Astro Seed

Tempe, AZ, USA

Product Development Intern

July 2023 - December 2023

- Developed and optimized a modular test bed for lettuce growth in space environment, reducing deployment time by 50%
- Designed a wooden frame with adjustable steel rails for flexible spacing using SolidWorks
- Implemented GD&T principles like Datums and Position Tolerance while modeling the Test Bed
- Initiated an ECR to enhance durability by replacing crack-prone 3D printed parts with steel
- Collaborated with bio-engineering team to identify critical physical and environmental parameters for lettuce growth

STAR LABS

Surat, GJ, India

Mechanical Intern

January 2022 - February 2022

- Utilizing 40x40mm Aluminum Extrusions, developed a static test bed to measure static thrust from Rockets (SolidWorks)
- Applied constraints and forces over the model and simulated the stability of system under maximum predicted thrust (SolidWorks)
- Designed a spring-loaded, servo-assisted electro-mechanical recovery mechanism with integrated parachute.
- Implemented Arduino-based control system utilizing altimeter for precise deployment.
- Defended designs and received acclaim for the most versatile test bed for increasing modularity and ease of transport by 100%

PROJECTS & OUTSIDE EXPERIENCE

Arizona State University

Tempe, AZ, USA

Student Researcher

August 2023 - December 2023

- Modified NACA 63-415 airfoil profile using Ansys Adjoint Solver to achieve a 15% improvement in lift-to-drag ratio.
- Designed model via SolidWorks & 3D printed through PLA for Wind Tunnel tests.

SASTRA university

Thanjavur, TN, India

Student Researcher

June 2019 - December 2019

- Fabricated an 8mm thick mild-steel nozzle & acrylic-glass test section for Mach 3 Wind Tunnel tests.
- Applied boundary layer theory for desired geometry, achieving a 3cm throat height.
- Used Schlieren imaging to validate shockwave and observe flow in line with calculations.

EDUCATION

Arizona State University

January 2022 - May 2024

Master's, Aerospace Engineering

SASTRA University

July 2016 - June 2020

Bachelor's, Aerospace Engineering